

GearDesign Report

Cylindrical gear(pair)

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1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	15.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	29.0000	61.0000
Facewidth (mm)	b	33.0000	33.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	93.1750	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	60.0460	126.3040
Tip diameter (mm)	d_a	64.046 (64.046 / 63.946)	130.304 (130.304 / 130.204)
Root diameter (mm)	d_f	55.046 (54.854 / 54.744)	121.304 (121.043 / 120.905)
Number of teeth spanned	k	4.0000	8.0000
Base tangent length (no backlash) (mm)	W_k	21.562 (21.496 / 21.458)	46.168 (46.079 / 46.032)
Effective Diameter of ball/pin (mm)	D_M	3.5000	3.5000
Diametral measurement over two balls (mm)	M_{dk}	64.938 (64.771 / 64.674)	131.300 (131.057 / 130.928)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.1473		1.1738		1.0000
Bending S.F	2.1491		2.1882		1.4000
Scuffing S.F (Flash Temp.)	2.7403				2.0000
Micropitting S.F	0.9634				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	1.1000	0.0000	0.5000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.5601
Overlap Ratio	1.3593
Total Contact Ratio	2.9195
Worst Pitch Line Velocity [m/s]	31.4400
Total Efficiency [%]	98.9329
Worst Mesh Loss [W]	593.7174
Worst Churning Loss [W]	1006.9725
Worst Peak-to-Peak T.E. [μm]	0.7954
Gear Total Mass [kg]	1.1897
Hunting tooth	Complete

1.5. Summary - Load cases

	Gear	Time [Hr]	Temperautre [°C]	Power [kW]	Speed [r/min]	Torque [N.m]	Contact SF (Left)	Bending SF (Left)	Scuffing SF (Left)	Contact SF (Right)	Bending SF (Right)	Scuffing SF (Right)	K_Hβ (Left)	K_Hβ (Right)	Mesh Loss (Left) [kW]	Mesh Loss (Right) [kW]	Churning Loss [kW]	PPTe (Left) [μm]	PPTe (Right) [μm]
LC	Gear1	10000.0000	80.0000	150.0000	10000.0000	143.2390	1.1473	2.1491	2.7403				1.3000		0.5937		0.1425	0.7954	
1	Gear2			-150.0000	-4754.0980	301.2970	1.1738	2.1882									0.8645		

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	15.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	29.0000	61.0000
Facewidth (mm)	b	33.0000	33.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	93.1749	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9500	0.9500
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.1034	
Transverse module (mm)	m_t	2.0706	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	20.6469	
Working transverse pressure angle (deg)	α_{wt}	20.6469	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	20.6648 / 20.6289	
Base helix angle (deg)	β_b	14.0761	
Helix angle at operating pitch circle (deg)	β_w	15.0000	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Cumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0725 / 0.1139	0.0984 / 0.1501
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0893 / 0.1363
Cumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1625 / 0.2723	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1475 / 0.2471	
Angluar backlash, min (deg)	j_θ, min	0.3102	0.1475
Angluar backlash, max (deg)	j_θ, max	0.5196	0.2470
Effective facewidth (mm)	b_{eff}	33.0000	
Sum of the profile shift coefficients	Sum_x	-0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0481 / -0.0756	-0.0653 / -0.0996
Prevent undercut profile shift coefficient	x_{Emin}	-0.8664	-2.9259
Max. profile shift coeff. (min top land)	x_{Emax}	1.3703	2.2479
Reference center distance, $x_1=x_2=0$ (mm)	a_d	93.1749	
Working center distance (mm)	a_w	93.1749	
Reference diameter (mm)	d	60.0460	126.3037
Operating pitch diameter (mm)	d_w	60.0460	126.3037
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	60.0531 / 60.0389	126.3186 / 126.2888
27		64.0460	130.3037
Tip diameter with tolerance (mm)	$d_a, e/i$	64.0460 / 63.9460	130.3037 / 130.2037

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diameter (mm)	d _f	55.0460	121.3037
Root diameter with tolerance (mm)	d _f , e/i	54.8537 / 54.7438	121.0427 / 120.9053
Tip form diameter (mm)	d _{Fa}	63.8460	130.1037
Tip form diameter with tolerance (mm)	d _{Fa} , e/i	63.6331 / 63.5119	129.8281 / 129.6835
Root form diameter (mm)	d _{Ff}	57.0425	122.7636
Root form diameter with tolerance (mm)	d _{Ff} , e/i	56.9511 / 56.9011	122.5656 / 122.4630
Active tip diameter (EAP) (mm)	d _{Na}	63.8460	130.1037
Active tip diameter with tolerance (mm)	d _{Na} , e/i	63.6331 / 63.5119	129.8281 / 129.6835
Active root diameter (SAP) (mm)	d _{Nf}	57.3188	123.3767
Active root diameter with tolerance (mm)	d _{Nf} , e/i	57.5410 / 57.4404	123.6005 / 123.4887
Mean diameter for mass calculation (mm)	d _m	59.3974	125.6138
Gear mass (kg)	m	0.3500	0.8396
Gear inertia (kg.m ²)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5000	0.5000
Effective tip clearance, e/i (mm)	c, e/i	0.7602 / 0.6195	0.7121 / 0.5852
Min. margin of involute contact, (d _{Nf_i} - d _{Ff_e}) / 2 (mm)	c _{Nff}	0.2447	0.4616

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	4.5000	4.5000
Tooth depth, e/i (mm)	h, e/i	4.6511 / 4.5462	4.6992 / 4.5805
Working depth (mm)	h _w	4.0000	
Addendum (mm)	h _a	2.0000	2.0000
Addendum, e/i (mm)	h _a , e/i	2.0000 / 1.9500	2.0000 / 1.9500
Dedendum (mm)	h _f	2.5000	2.5000
Dedendum, e/i (mm)	h _f , e/i	2.5962 / 2.6511	2.6305 / 2.6992
Number of virtual gear teeth	z _n	31.9106	67.1222
Tooth thickness coeff.	s _{nc}	1.5708	1.5708
Transverse tooth thickness (mm)	s _t	3.2524	3.2524
Normal tooth thickness (mm)	s _n	3.1416	3.1416
Normal tooth thickness at d _a (mm)	s _{an}	1.4876	1.5834
Normal tooth thickness at d _a , e/i (mm)	s _{an} , e/i	1.4636 / 1.3708	1.5292 / 1.4341
Normal tooth thickness at d _{Fa} (mm)	s _{Fan}	1.5877	1.6702
Normal tooth thickness at d _{Fa} , e/i (mm)	s _{Fan} , e/i	1.6772 / 1.5762	1.7522 / 1.6395
Normal space width at d _f (mm)	e _{fn}	0.0000	1.5600
Normal space width at d _f , e/i (mm)	e _{fn} , e/i	0.0000 / 0.0000	1.5911 / 1.6083

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	6.2832	
Transverse pitch (mm)	p_t	6.5048	
Normal base pitch (mm)	p_bn	5.9043	
Transverse base pitch (mm)	p_bt	6.0870	
Axial pitch (mm)	p_x	24.2764	
Lead (mm)	p_z	704.0146	1480.8582

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	9.4966	
Length of path of contact, e/i (mm)	g_α, e/i	8.9710 / 8.6038	
Length of approach POC of pinion (A to C) (mm)	g_f1	4.9249	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	4.6147 / 4.5724	
Length of recess POC of pinion (C to E) (mm)	g_a1	4.5718	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	4.3563 / 4.3362	
Length T1 - A (mm)	T1A	5.6615	
T1A tolerance (mm)	e/m/i	6.1992 / 6.0804 / 5.9616	
Length T1 - B (mm)	T1B	9.0711	
T1B tolerance (mm)	e/m/i	8.8456 / 8.7808 / 8.7160	
Length T1 - C (mm)	T1C	10.5863	
T1C tolerance (mm)	e/m/i	10.5964 / 10.5863 / 10.5763	
Length T1 - D (mm)	T1D	11.7485	
T1D tolerance (mm)	e/m/i	12.2862 / 12.1674 / 12.0486	
Length T1 - E (mm)	T1E	15.1581	
T1E tolerance (mm)	e/m/i	14.9326 / 14.8678 / 14.8030	
Length T2 - A (mm)	T2A	27.1927	
T2A tolerance (mm)	e/m/i	26.8614 / 26.7738 / 26.6861	
Length T2 - B (mm)	T2B	23.7831	
T2B tolerance (mm)	e/m/i	24.1694 / 24.0734 / 23.9774	
Length T2 - C (mm)	T2C	22.2678	
T2C tolerance (mm)	e/m/i	22.2890 / 22.2678 / 22.2467	
Length T2 - D (mm)	T2D	21.1057	
T2D tolerance (mm)	e/m/i	20.7744 / 20.6867 / 20.5991	
Length T2 - E (mm)	T2E	17.6961	
T2E tolerance (mm)	e/m/i	18.0823 / 17.9864 / 17.8904	
Length T1 - T2 (mm)	T1T2	32.8542	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	32.8854 / 32.8542 / 32.8230	
Diameter of single contact point A (mm)	d_A	57.3188	130.1037
Diameter of single contact point B (mm)	d_B	59.0456	127.4039
Diameter of single contact point C (mm)	d_C	60.0460	126.3037
Diameter of single contact point D (mm)	d_D	60.9044	125.5030
Diameter of single contact point E (mm)	d_E	63.8460	123.3767
Transverse contact ratio	ϵ_α	1.5601	
Overlap ratio	ϵ_β	1.3593	
Total contact ratio	ϵ_Y	2.9195	
Total contact ratio with tolerance	e/m/i	2.833 / 2.803 / 2.773	
Specific sliding at Tip	ζ_a	0.4450	0.5621
Specific sliding at Root	ζ_f	-1.2835	-0.8018

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	8.0000	8.0000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	18.0000	19.0000
Profile form deviation (μm)	f_fa	8.5000	8.5000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	10.0000	10.0000
Helix slope deviation (μm)	f_Hβ	9.0000	9.5000
Total helix deviation (μm)	F_β	13.0000	14.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	28.0000
Runout (μm)	F_r	22.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.0000	8.0000
Single flank composite, total (μm)	F_isT	33.0000	36.0000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	37.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	8.0000
Base tangent length (mm)	W_k	21.5617	46.1682
Base tangent length with tolerance (mm)	W_k, e/i	21.4959 / 21.4583	46.0790 / 46.0320
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0893 / -0.1363
Diameter of contact point (mm)	d_M	59.9268	126.3520
Theoretical diameter of ball/pin (mm)	D_M.theory	3.4057	3.3720
Effective Diameter of ball/pin (mm)	D_M.eff	3.5000	3.5000
Radial single ball measurement (mm)	M_rk	32.5142	65.6710
Radial single ball measurement with tolerance (mm)	M_rk, e/i	32.4304 / 32.3820	65.5496 / 65.4852
Diametral two ball measurement (mm)	M_dk	64.9382	131.2997
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	64.7708 / 64.6741	131.0570 / 130.9281
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1673 / -0.2640	-0.2427 / -0.3716

3. Detail rating

3.1. General

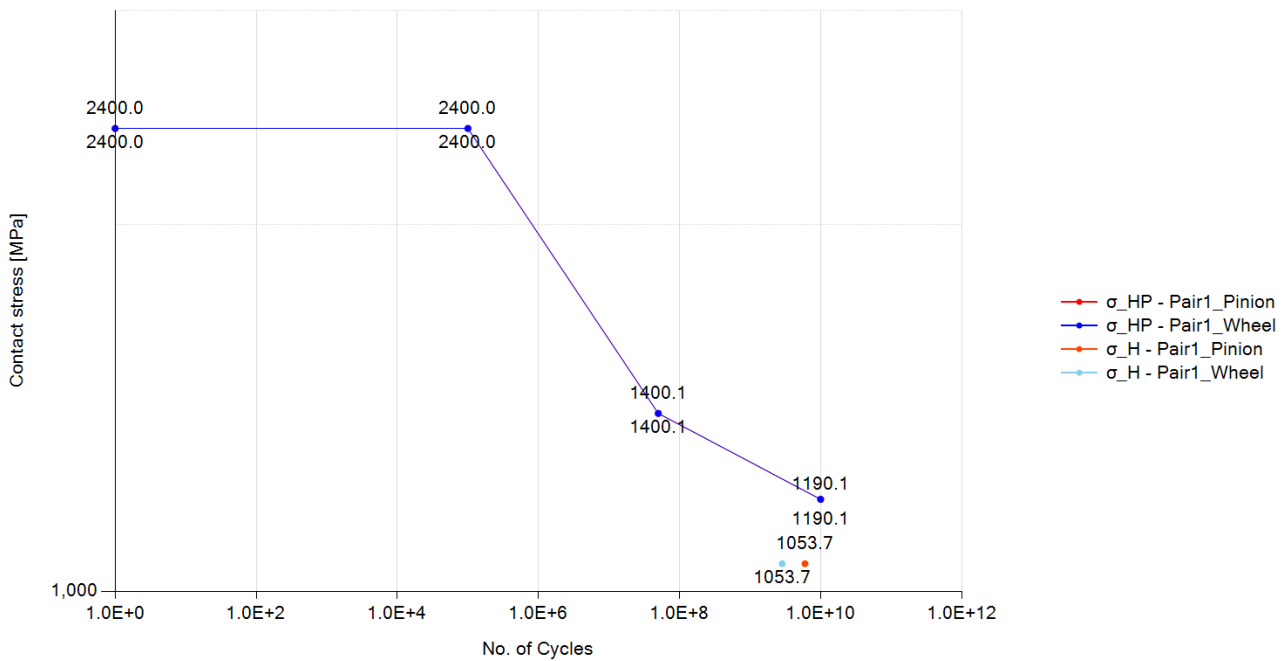
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4754.0980
Nominal tangential load at d (N)	F_t	4770.9890	
Nominal axial load at d (N)	F_a	1278.3830	
Nominal radial load at d (N)	F_r	1797.7550	
Nominal normal load at d (N)	F_n	5256.2840	
Nominal tangential load per unit facewidth (N/mm)	w	144.5750	
Circumferential speed at d (m/s)	v	31.4400	
Sliding velocity at Tip dia. (m/s)	v_g	7.0640	7.6090
Nominal tangential load at d_w (N)	F_{tw}	4770.9890	
Nominal axial load at d_w (N)	F_{aw}	1278.3830	
Nominal radial load at d_w (N)	F_{rw}	1797.7550	
Circumferential speed at d_w (m/s)	v_w	31.4400	
Running-in value (μm)	y_p	0.6000	
Running-in value (μm)	y_f	0.6370	
Effective single pitch & profile deviation (μm)	f_{pb_eff}	7.4000	
Effective single pitch & profile deviation (μm)	f_{fa_eff}	7.8620	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/ μm)	c'_{th}	17.8740	
Single tooth stiffness (N/mm/ μm)	c'	12.0930	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/ μm)	$c_{y\alpha}$	17.1740	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/ μm)	$c_{y\beta}$	14.5980	
Mean diameter for K_v (mm)	$d_m (K_V)$	59.5460	125.8040
Moment of inertia for K_v ($\text{kg}\cdot\text{mm}^2$)	$J (K_V)$	280.6560	3682.3960
Moment of gear inertia per unit facewidth ($\text{kg}\cdot\text{mm}$)	J^*	8.5050	111.5880
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0108	0.0319
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0081	
Resonance running speed (rpm)	n_{E1}	15202.0710	
Resonance ratio	N	0.6580	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.2260	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2540	
Transverse load factor - flank	$K_{H\alpha}$	1.1160	
Transverse load factor - root	$K_{F\alpha}$	1.1160	
Number of load cycles	N_L	6.0000×10^9	2.8520×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



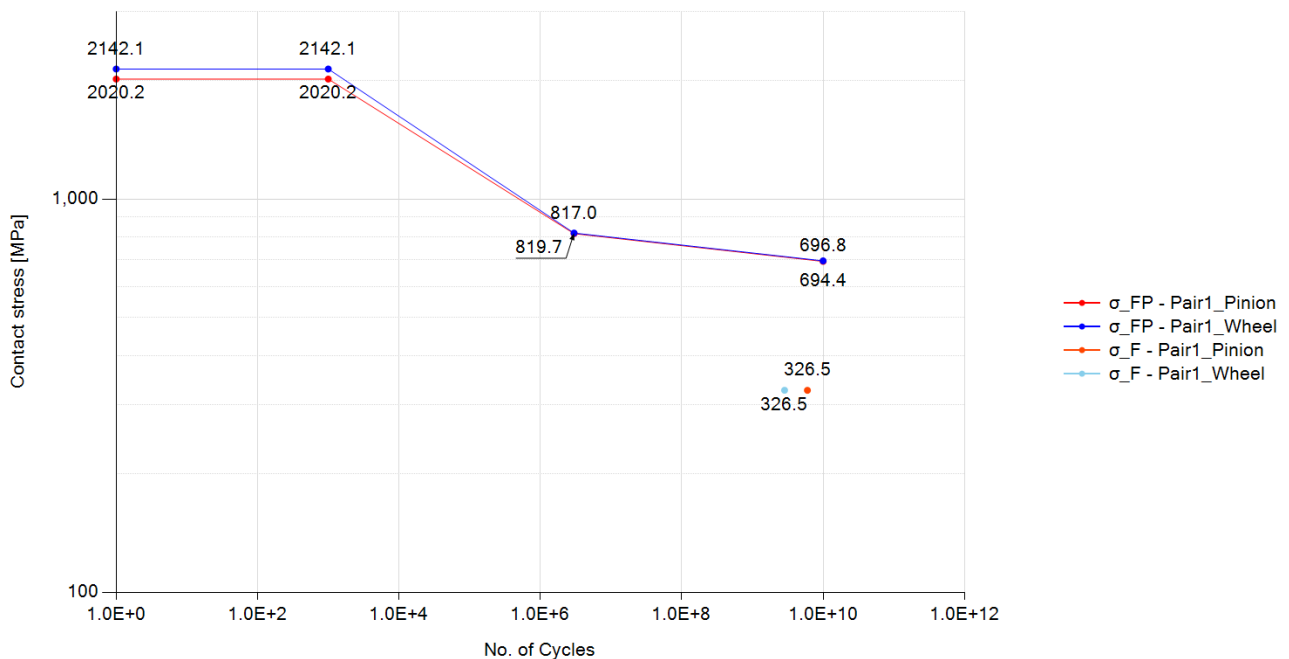
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_{ϵ}	0.8010	
Zone factor	Z_H	2.4250	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_{β}	1.0170	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm^2)	σ_{H0}	706.6340	
Contact stress (N/mm^2)	σ_H	1053.6840	1053.6840
Life factor	Z_{NT}	0.8630	0.8830

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0340	1.0340
Roughness factor	Z_R	0.9550	0.9550
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1400.1480	1400.1480
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1190.1260	1190.1260
Permissible contact stress (N/mm ²)	σ_{HP}	1208.9210	1236.8110
Safety factor for Contact	S_H	1.1470	1.1740
Contact damage (%)	-	1.1330	0.5380
Cycle to failure	-	5.2970×10^{11}	5.2970×10^{11}

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.1640	1.0890
Stress correction factor	Y_S	1.8630	1.9920

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.8670	19.8100
Bending moment arm (mm)	h_{Fe}	2.0840	2.2420
Tooth thickness at root (mm)	s_{Fn}	4.0980	4.3820
Tooth root radius (mm)	ρ_F	1.1390	1.0340
Helix angle factor	Y_β	0.9710	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	152.2470	152.2800
Tooth root stress (N/mm ²)	σ_F	326.4610	326.5320
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9930	0.9960
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	2020.2400	2142.0580
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	816.9870	819.7340
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	694.4390	696.7740
Permissible tooth root stress (N/mm ²)	σ_{FP}	701.5830	714.5070
Safety factor for Bending	S_F	2.1490	2.1880
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	1.1690×10^{19}	1.3670×10^{19}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	28.0500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	321.4590	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	10.5230	
Angle factor	X_αβ	0.9730	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0470	
Maximum flash temperature (deg)	θ_fl_max	86.9080	
Mean flash temperature (deg)	θ_fl_mean	30.2050	
Total mass temperature (deg)	θ_M	97.0360	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	183.9440	
Safety factor for flash temperature	S_B	2.7400	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.1220
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.2033
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.9634
transverse tangential load at reference cylinder per mesh (N)	F_t	4770.9890
transverse load in plane of action (N)	F_bt	5098.4570
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1740
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_Bγ	1.2770
load losses factor	H_v	0.1090
mean coefficient of friction	μ_m	0.0570
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.8330
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	19.4670
sliding velocity at point A (m/s)	v_g,A	-7.6090
tangential velocity on pinion at point A (m/s)	v_r1,A	5.9290
tangential velocity on wheel at point A (m/s)	v_r2,A	13.5380
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	101.5230
flash temperature at point A (°C)	θ_fl,A	104.7110
contact temperature at point A (°C)	θ_B,A	206.2330
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3220×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	7.6790×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	1.7950
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0010
normal radius of relative curvature at point A (mm)	ρ_n,A	4.8310
material parameter	G_M	2991.7450
local velocity parameter at point A	U_A	6.2330×10 ⁻¹¹
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.1120
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	979.8500
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1461.0830

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.1034
Tangential speed at working pitch dia. (m/s)	v_{wt}	31.4400
Tangential force at pitch dia. (N)	F_t	4770.9891
Tooth normal force, transverse section (N)	F_{bt}	5098.4566
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	22.1720
Equivalent radius of curvature (mm)	ρ	7.3973
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	154.4987
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0365
Tooth loss factor	H_v	0.1085
Mesh power loss (W)	P_{VZP}	593.7174
Mesh efficiency (%)	η_z	99.6042

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4754.0984
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	30.0230	63.1518
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	69818.7772	69818.7772
Froude number	Fr	3356.1584	1595.5507
Centrifugal acceleration (m/s ²)	γ	13773.8903	3988.7507

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	64.0460	130.3037
Pitch diameter of gear (mm)	d_p	60.0460	126.3037
Facewidth (mm)	b	33.0000	33.0000
Dimensionless drag torque	C_m	0.0010	0.0022
Immersion area of the gear (m^2)	S_m	0.0107	0.0291
Total oil churning loss (W)	P	142.4616	864.5109
Total oil churning torque (N.m)	T	0.1360	1.7365

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000